

Measurements of fast ions and neutrons emitted from PF-1000 plasma focus device

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Abstract

The paper presents the results of recent experiments performed within the PF-1000 facility. The main aim of these studies was to determine the neutron emission characteristics, which could enable an approximate settlement of the problem of the neutron emission mechanism. The total neutron yield and angular distribution anisotropy were measured with four silver-activation counters placed at different angles to the electrode axis. The neutron energy spectra were determined by means of an 85 m upstream positioned scintillation probe. Fast ion beams were investigated with solid-state nuclear track detectors (SSNTD) of the CR-39 type covered with Al-foil filters and placed at a semicircular support around the PF-1000 electrode outlet.

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Keywords: Plasma focus; Neutrons yield; Neutrons spectra; Angular anisotropy; Fast ion beams

1. Introduction

The problem of a mechanism of the neutron emission from pinched plasma discharges (Z-pinch, X-pinch, Plasma-Focus, etc.) is very intriguing, and it still remains unclear. In spite of many efforts, both theoretical and experimental studies to solve the problem, the prevailing mechanism of the interactions between two nuclei of deuterons has not been found. Therefore, the recent experi-

ments with the PF-1000 facility were aimed at the investigation of the neutron emission characteristics, which could enable an approximate settlement of the problem of the neutron emission to be achieved. Intensity, angular distribution, and energy spectra of nuclear reaction neutrons, i.e. characteristics which were measured in our experiments deliver such data as well as some information on heating and accelerating processes of the ions.

The other main aim of the experiments was to define characteristic features of the investigated discharges. The problem was to find the instances

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when the current sheath layer has been collapsed and the maximum compression has been achieved, as well as the phases when the hard X-ray and neutron pulses have been emitted.

Assuming that fast ion beams contribute essentially to the total neutron yield, direct measurements of the fast ions emission were also performed.

2. Experimental

The paper presents the results of experiments performed within the PF-1000 facility, which was equipped with a new set of the large electrodes and operated at discharge energy level ranging up to 1 MJ. The set consisted of two coaxial electrodes and an alumina insulator. The inner electrode (anode) was made of copper, and it was 447 mm in length and 244 mm in diameter. The insulator, of the same diameter and 113 mm in length, embraced the inner electrode at its basis. The outer electrode (cathode) was composed of 24 stainless-steel rods, each of 32 mm in diameter and 600 mm in length. They were distributed symmetrically upon the cylinder surface of 368 mm in diameter. The condenser bank of capacitance 1332 μF was charged to voltage ranging 20–40 kV, which corresponded to discharge energies ranging from 266 to 1064 kJ.

The total neutron yield and neutron emission angular distributions were investigated using four silver activation counters, which were positioned, at different angles to the main electrode axis (Fig.

1). Simultaneously, time-resolved measurements of hard X-ray and neutron pulses were carried out by means of four scintillation probes, which were located at different distances from the electrode outlet (6.5 m downstream, 40 and 85 m upstream) (Fig. 1). Fast ions were investigated by means of solid-state nuclear track detectors of the CR-39 type, which were placed at different angles to the axis, on a semicircular support at a distance of 550 mm from the electrode end. Some track detector samples were partially covered with Al-foil of different thickness. Such an arrangement made it possible to measure the angular distribution of different energy ions.

In the described experiments additional diagnostics were also applied, e.g. a Rogovski coil and a small magnetic (current derivative) probe were used to monitor the voltage and discharge current. Soft X-ray and “visible” plasma radiation of the moving plasma sheath were detected by means of PIN diodes covered with different foil filters. Current sheath dynamics was investigated by means of a two-frame camera, which was placed side-on the discharge axis.

3. Results and discussion

The PF-1000 facility has been operated for a few years, gradually at increasing discharge energy, and the obtained results have been successively published [1–5]. The main efforts of the experiments were however directed towards the optimization of the machine and the determination of neutron emission characteristics. The results obtained so far can be summarized as follows: (1) The PF-1000 facility can be operated at discharge energy levels exceeding 1 MJ, with a regular and reproducible neutron emission (of the order of 10^{10} – 10^{11} neutrons/shot); (2) It was found that for each filling pressure value (p_0) neutron yield (Y_n) increases initially as a function of the charging voltage (V_0) up to a maximum, and after that it decreases with a further increase in V_0 ; (3) The neutron emission anisotropy coefficient, defined as $Y_n(30^\circ)/Y_n(90^\circ)$ decreases with an increase in the filling pressure value (from 1.4 at 133 Pa, to less than 1.2 at 665 Pa); (4) Two (or three) neutron

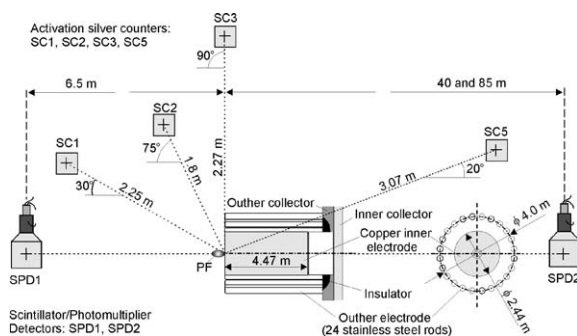


Fig. 1. General experimental set-up also showing the position of the neutron activation counters and scintillation probes.

pulses, about 2 μs apart, were observed in most of the discharges; (5) The maximum amplitude of the neutron energy spectra, which were measured in the upstream direction was shifted towards lower energy values, i.e. to 2.2–2.3 MeV.

Fig. 2 presents typical time sequences of the characteristics events, as determined for a PF-1000 discharge.

The first soft X-ray pulse, which was recorded with the PIN diode, reveals a fine structure, and the first spike of this pulse coincides with the signals from the dI/dt probe. The first hard X-ray pulse from the scintillation probe, which is only shifted (in Fig. 2) by X-ray time-of-flight, is also synchronized with this instance. The neutron pulses, which were recorded with the scintillation probes, show characteristic pre-pulses, which prove that the neutron emission starts earlier than the X-ray radiation.

Taking as a reference for the $\Delta t = 0$ emission time the maxima of the corresponding X-ray pulses, we tried to calculate the neutron energy from the measured time-of-flight and to convert the neutron pulses into a neutron energy spectrum (Fig. 3).

One can see that the obtained spectrum shows the maximum at the energy 2.2–2.3 MeV, i.e. at the value lower than 2.45 MeV, which follows from the D–D reaction energy balance.

Fig. 4 presents pictures of typical ion tracks, which have been registered at two different angles to the electrode axis, and on detector samples covered with different Al-foil filters.

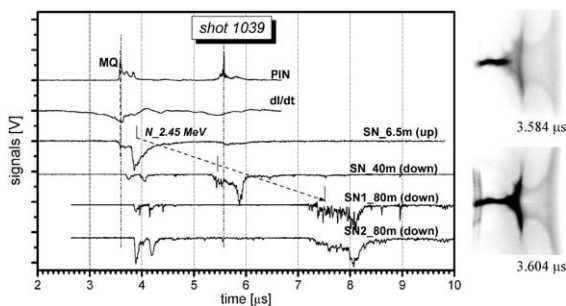


Fig. 2. Typical time sequence of the characteristic events determined for a PF-1000 discharge: soft X-ray, dI/dt , hard X-ray and neutron pulses.

These samples were irradiated by fast deuterons emitted from a single PF-1000 shot, which was performed at the D_2 -filling pressure equal to 252.7 Pa. After the irradiation these samples were etched under standard conditions and scanned with an optical microscope. The optical analysis has shown the ion crater densities ranging from 10^3 to 10^5 craters/ mm^2 up to a saturation level, as presented in Figs 4 and 5.

Taking into account energy losses of D^+ -ions in Al (D^+ -ions of energy > 250 keV penetrate through about 1.5- μm Al-foil, and 500 keV ions through 4- μm Al-foil), and the detection characteristics of the used detector [6,7], one could estimate that the uncovered detector samples recorded D^+ -ions of energy > 80 keV, samples covered with 1.5 μm Al-foil registered D^+ -ions faster than 330 keV, and samples masked with 4 μm Al-foil revealed tracks of ions of energy > 580 keV. It was, however, observed

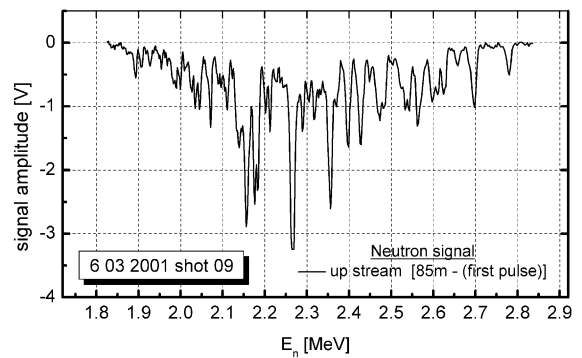


Fig. 3. Neutron energy spectrum of the first pulse of shot 0603-09.

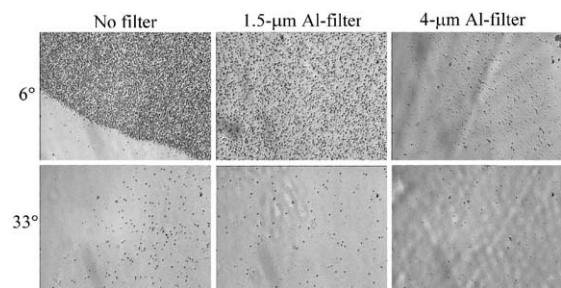


Fig. 4. Examples of ion tracks, which were recorded upon SSNTDs without and with Al-filters, at different angles to the z -axis.

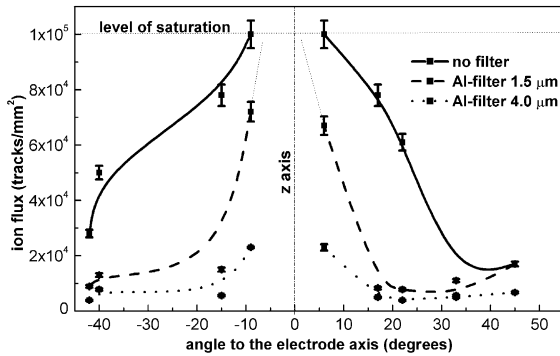


Fig. 5. Angular distributions of fast deuteron.

that the fast ions emission is not reproducible from shot to shot, and decreases with an increase in the filling pressure. Angular distributions of different energy primary deuterons, as obtained for a single PF-1000 shot performed at the D_2 —filling pressure equal to 252.7 Pa are presented in Fig. 5.

One can see that the PF-1000 facility generates relatively intensive fluxes of fast ions (10^5 ions/ mm^2) of energy above several dozen keV and that these fluxes are largely emitted downstream along the electrode axis direction.

4. Conclusions

The PF-1000 facility can be operated at discharge energy levels exceeding 1 MJ, with a regular and reproducible neutron emission of the order of 10^{10} – 10^{11} neutrons/shot.

For each filling pressure the value of the neutron yield increases initially as a function of the charging voltage up to a maximum, and after that it decreases with a further increase in V_0 .

The neutron emission anisotropy coefficient decreases with an increase in the filling pressure (from 1.4 at 133 Pa to less than 1.2 at 665 Pa).

The neutron energy spectra, which were measured in the upstream direction were shifted toward lower energy values, i.e. to 2.2–2.3 MeV.

There were mostly two (or three) neutron pulses (about 2 μs apart) recorded. Intensive ion beams of energy above 100 keV were recorded. Most of these beams are emitted downstream along the electrode axis.

Acknowledgements

This work has been supported by IAEA CRP Grants No. 11940, 11941 and Grant of the European Commission G4M-CT-2002-04037.

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